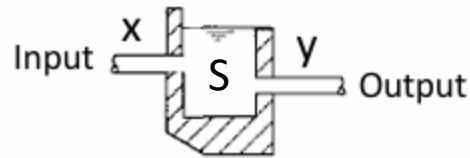


What is memory?

In its simplest definition, “**Memory**” is something which is able to **store the previous states**.

In other words, “**Memory**” enables the existence of “**State Space**”.

In order to understand what really the **state space** is, we could look at the following water tank:



The Output Flow **Y** is dependent on the Input Flow **X**, as well as the Capacity of Storage **S**, which is actually able to define the **State Space**.

$$Y = F(X, S)$$

In such case, we could loosely define the Water Tank as the **Memory of the Flow** of water.

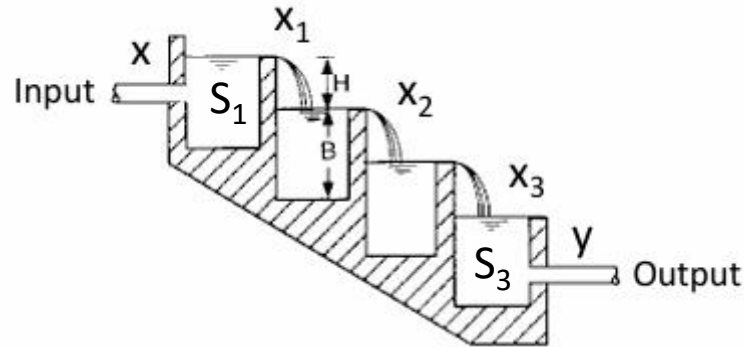
Since the output **Y** is dependent on both **X** and **S**, let us define a **Deterministic Causality**, such that:

If **X**, then **Y**, given **S**

If we know the State **S** and Input **X**, then we can **definitely determine** the Output **Y**.

State Space

Now, let us look at a more interesting State Space, in which we have Cascading Storage.



In such case, we can define the “**Chain of Deterministic Causality**”, such that:

If X , then X_1 , given S_1

If X_1 , then X_2 , given S_2

...

If X_n , then Y , given S_n

Thus, the State Space $\{S_1, \dots, S_n\}$ defines the **Deterministic State Space**, which can be “**Inference**” or “**Reason**” the chain of causality from X to Y .

Deterministic Causality

Given a State Space $\{S_1, \dots, S_n\}$ (Memory), the Deterministic Causality is defined in such a way that:

If **X**, then **Y**, given $\{S_1, \dots, S_n\}$

Inversely:

If **Y**, then **X**, given $\{S_1, \dots, S_n\}$

As we can see, the Memory or State Space is something which can connect the past events to the present events, through **Causality**.

More importantly, it is such Deterministic Causality that defines the “**Rational Determinism**”, or rather “**Rationality**”, which is defined as **Aristotelian Logic**.

If **X**, then **Y**, given $\{S_1, \dots, S_n\}$

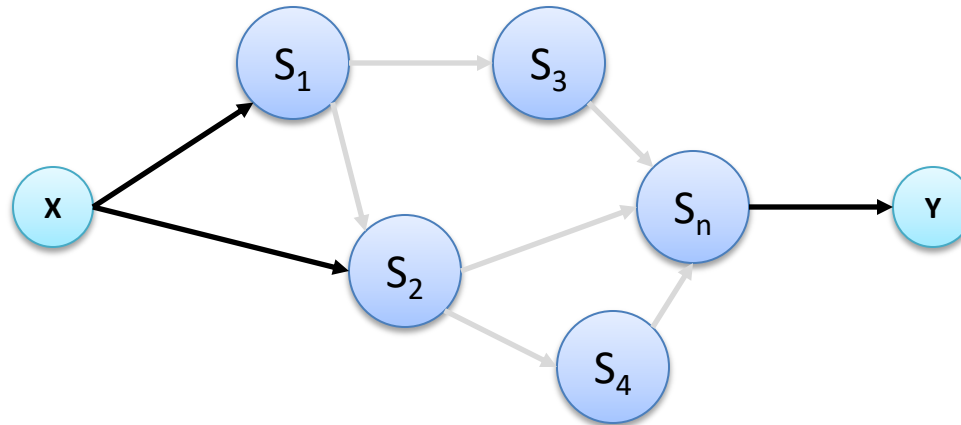
Given State Space $\{S_1, \dots, S_n\}$, if X happens, then Y must definitely happen, without exception.

As we can see Deterministic Causality has no room for “**Uncertainty**”.

Deterministic Reasoning

If **X**, then what is **Y**, given $\{S_1, \dots, S_n\}$

With Deterministic Causality, **Deterministic Reasoning** is possible. What is Deterministic Reasoning is reaching the **Definite Output State**, given **State Space** and **Input State**.



Given Yangon city is 60 km away from Bago City (**S**), and if we are driving with a speed of 60 km per hour (**X**), then we can travel from Yangon to Bago in one hour (**Y**).

As we can see, **Deterministic Reasoning** will always lead to **Definite Output State**, which, however, leads to **Optimistic Reasoning**, which does not take account for any **Uncertainty**.

It is why **Deterministic Rationalism** always results in **Idealism**, which has no Dubious States.

Non Deterministic Causality

However, in Reality, it is not always possible to reach a Definite Output **Y** from an Input **X**, given a state space $\{S_1, \dots, S_n\}$.

Sometimes, more than an Output **Y** is possible from an Input **X**, given a state space $\{S_1, \dots, S_n\}$.

Thus, **Non Deterministic Causality** is defined such that:

If **X**, then $\{Y_1, \dots, Y_m\}$ given $\{S_1, \dots, S_n\}$

In other words, given a State Space $\{S_1, \dots, S_n\}$, an Input **X** could probably result in many possible Outputs $\{Y_1, \dots, Y_m\}$.

However, formally, it is really quite difficult to define **Non Deterministic Causality**, which is why **Stochastic Causality** is defined, along with **Probability**.

Actually, **Non Deterministic Causality** is more aligned with Reality, which results in **Realism**, which allows Dubious States.

Stochastic Causality

Stochastic Causality, with **Probability**, is defined such that:

If $P(\mathbf{X})$, then $P(\mathbf{Y})$, given $\{\mathbf{S}_1, \dots, \mathbf{S}_n\}$

Given a State Space $\{\mathbf{S}_1, \dots, \mathbf{S}_n\}$, an Input $\mathbf{X}$, with a Probability of $P(\mathbf{X})$, could result in Output $\mathbf{Y}$, with a Probability of $P(\mathbf{Y})$.

Actually, we can define **Stochastic Causality** a bit differently, with **Bayes Theorem**, such that:

$$P(\mathbf{Y} \mid \mathbf{X}, \mathbf{S})$$

However, with the assumption of **Conditional Independence**:

$$P(\mathbf{Y} \mid \mathbf{X}, \mathbf{S}) = P(\mathbf{Y} \mid \mathbf{S})$$

Then, $P(\mathbf{Y} \mid \mathbf{X}, \mathbf{S})$ varies with $P(\mathbf{Y}, \mathbf{X} \mid \mathbf{S})$ as $P(\mathbf{X} \mid \mathbf{S})$ will be constant.

$$P(\mathbf{Y}, \mathbf{X} \mid \mathbf{S}) = P(\mathbf{Y} \mid \mathbf{S}) P(\mathbf{X} \mid \mathbf{S})$$

$$P(\mathbf{Y}, \mathbf{X} \mid \mathbf{S}) = P(\mathbf{Y} \mid \mathbf{X}, \mathbf{S}) P(\mathbf{X} \mid \mathbf{S})$$

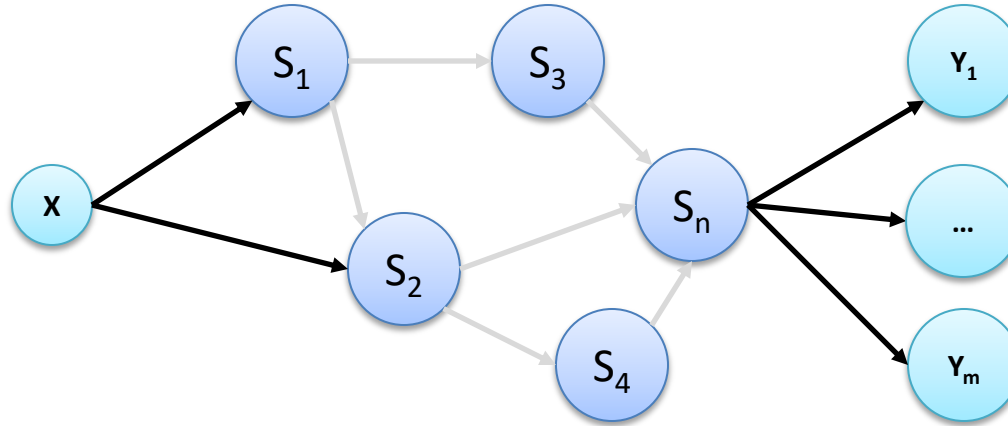
$$P(\mathbf{Y} \mid \mathbf{X}, \mathbf{S}) = P(\mathbf{Y}, \mathbf{X} \mid \mathbf{S}) / P(\mathbf{X} \mid \mathbf{S})$$

$$P(\mathbf{Y} \mid \mathbf{X}, \mathbf{S}) \propto P(\mathbf{Y}, \mathbf{X} \mid \mathbf{S}), \text{ if } P(\mathbf{X} \mid \mathbf{S}) = \text{Constant}$$

Stochastic Reasoning

If $P(\mathbf{X})$, what is $P(\mathbf{Y})$, given $\{S_1, \dots, S_n\}$

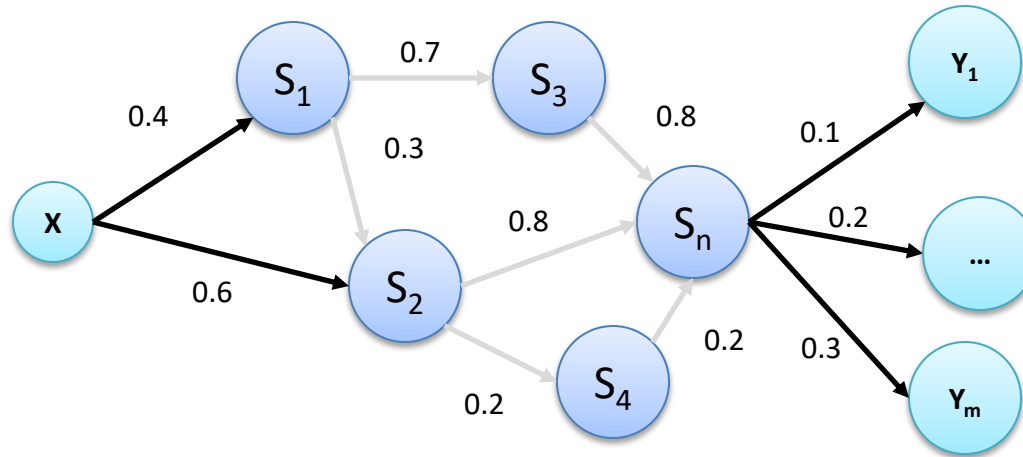
With Stochastic Causality, **Stochastic Reasoning** is reaching the **Probable Output States**, given the Probability of **State Space** and **Input State**.



Given Yangon city is 5 km away from the sea (**S**), and if we are in Monsoon Season (**X**), then what is the probability of rain in Yangon tomorrow (**Y**).

Bayesian Network

Bayesian Network is an **Acyclic Network** of the Chain of **Bayesian Probability**, which is generally used in **Stochastic Reasoning**, or more simply **Bayesian Inference**.



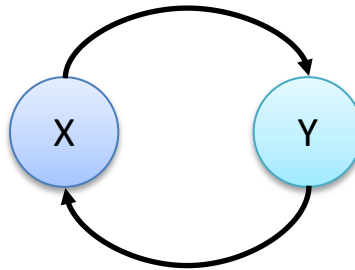
However, Bayesian Network also assumes **Conditional Independence**, which in turn forbids the Existence of **Loops** or **Cycles** in the network.

Therefore, Bayesian Network must be an **Acyclic Network** of **Chain of Conditional Probability**, without any Cyclic Dependency.

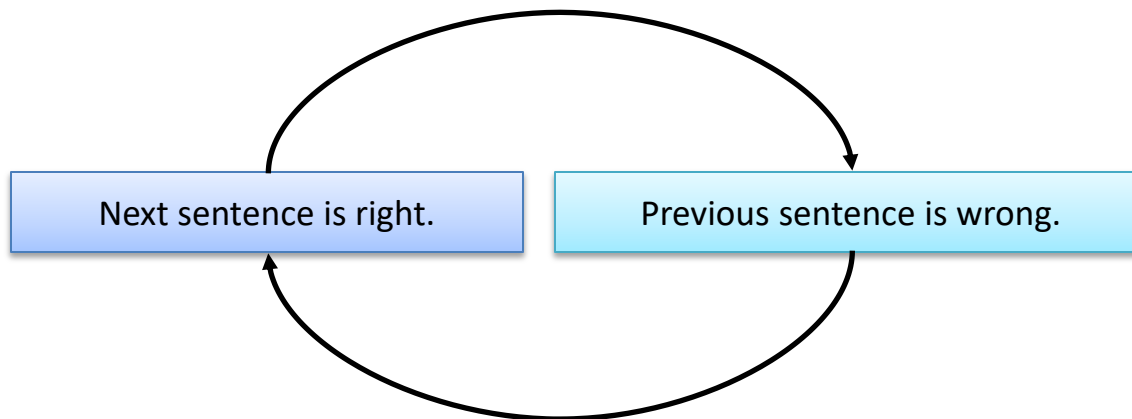
Reasoning , either Stochastic or Deterministic, is not perfect in that it cannot allow **Cyclic Inference**, which is interesting because most **Asian Philosophies** are based on Cycles.

Cyclic Dependency

Cyclic Dependency, or rather **Cyclic Inference** is not generally allowed in any form of Reasoning, either Stochastic or Deterministic, as it could never reach **Rational Conclusion**.



For example, what is the logical conclusion of which one is correct?



Stochastic Learning

Reasoning, either Deterministic or Stochastic, is reaching the Output **Y** from Input **X**, given the State Space **S**.

If $P(\mathbf{X})$, what is $P(\mathbf{Y})$, given $\{\mathbf{S}_1, \dots, \mathbf{S}_n\}$

Then, **Learning**, on the other hand, is:

If $P(\mathbf{X})$, what is $\{\mathbf{S}_1, \dots, \mathbf{S}_n\}$, given $P(\mathbf{Y})$

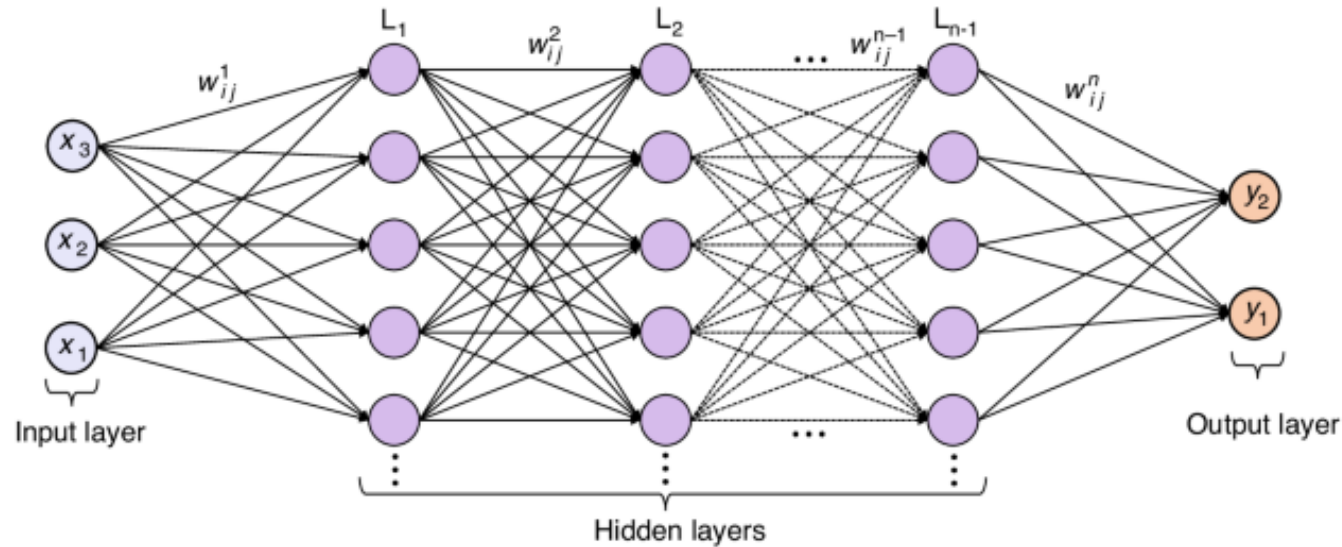
It turns out that Learning, especially **Stochastic Learning**, is far more difficult than **Stochastic Reasoning**.

More precisely, Stochastic Learning is updating the State Space $\{\mathbf{S}_1, \dots, \mathbf{S}_n\}$, based on Input **X** and Output **Y**, which is extremely hard due to the various possibilities of states.

Actually, **Stochastic Learning** is a **NP Hard** Problem, which is why Bayesian Network is mainly used for **Bayesian Inference**, rather than **Bayesian Learning**.

Later, Stochastic Learning is made possible by a Network of Associative Neurons, which is now known as **Artificial Neural Network (ANN)**.

Neural Network



Machine Learning, which is actually **Stochastic Learning**, is just updating the Parameters of the Network $\{\mathbf{W}_1, \dots, \mathbf{W}_n\}$, given Input $\mathbf{X}$ and Output $\mathbf{Y}$.

Such updates on parameters is usually done by “**Back Propagation**”, which employs **Feedback Deterministic Causality (Change)**, based on **Chain Rule of Derivative**.

Even then, updating millions, billions to trillions of parameters is not a small feat, which is why Training Neural Network is highly computationally expensive.

Predictive Coding

Though Neural Networks are immensely powerful in terms of **Stochastic Learning** and **Stochastic Reasoning**, their drawbacks are immensely huge as well.

First of all, **Stochastic Learning** of Neural Networks is extremely time consuming, which makes it impossible to learn in real-time.

Real-Time Learning would be extremely important if **Stochastic Reasoning** must be performed in Dynamic Environments, for which Real Time **Stochastic Learning** would be necessary.

It turns out that Our Memory in Humans, learn entirely differently.

Many neuroscientists suggest Our Memory could employ different **Stochastic Learning Mechanism**, known as “**Predictive Coding**”, instead of “**Back Propagation**”, which would enable us to learn in real time.

Even then, “**Learning**” is still not small feat for us; we would still have to spend at least 10 to 20 years to have a decent education.

At any rate, “**Learning**” is just updating our model in our memory. The complications comes from the very fact that our memory is the Complex network of Neurons.

What is memory?

“Memory” is something which is able to **store the states** in **State Space**.

However, updating such previous states is not small feat, especially if such state space is made up of interconnected network of various nodes.

Thus, learning and reasoning is highly associated with such interconnected network of state space.

At any rate, the exact model of our memory is still unknown and how we actually learn and reason is still mysterious.

What is more interesting our memory is also associated with emotions, which is also related to learning and reasoning.